

Machine Constraints & Standard Recipes in Textile Dyeing

Operational Limits, Process Calculations, Recipe Formulations, Thermal Curves & Quality Diagnostics

1. Executive Summary & Machine-Dye Interactions

Industrial textile wet processing requires precise synchronization between physical machine parameters (liquor circulation, nozzle hydraulics, differential pressure, rope speed) and chemical dye kinetics (exhaustion rate, migration, fixation chemistry). Deviation outside specified equipment operating envelopes directly results in unlevelness, fabric surface abrasion, batch-to-batch shade variation, or high re-dye costs. This engineering guide details standard constraints, calculations, recipes, and control curves for primary industrial substrates.

2. Primary Machine Types & Operating Constraint Parameters

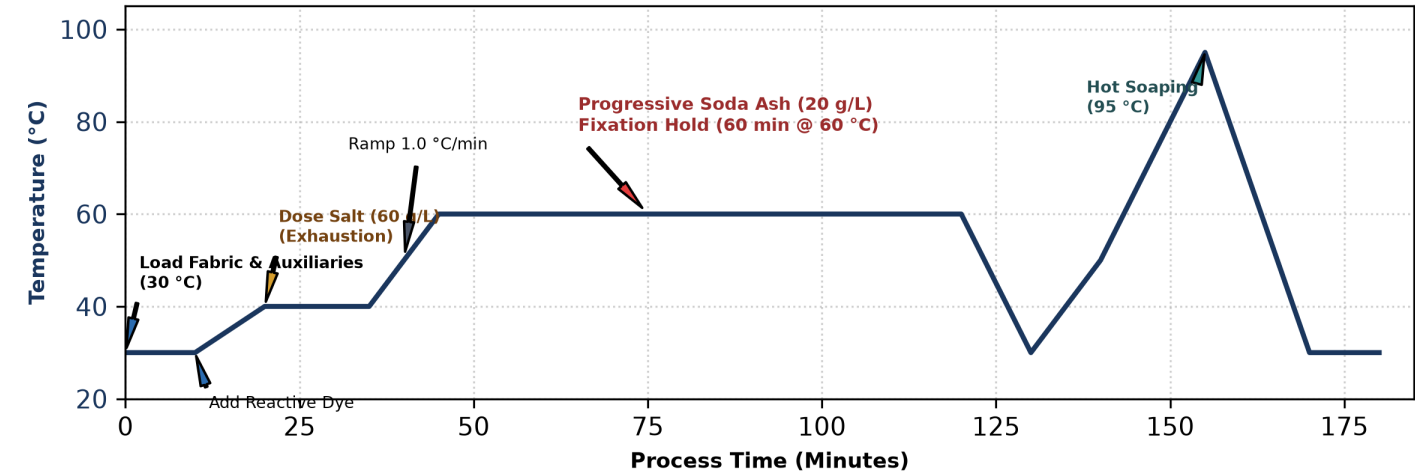
Machine Category	Target Substrate	Material-to-Liquor Ratio (MLR)	Critical Flow / Hydraulic Constraint	Ramp Rate & Thermal Limits	Primary Risk Factor
Jet / Soft-Flow Dyeing	Knits & Wovens (Rope form)	1:3.5 – 1:6 (Airflow) 1:6 – 1:10 (Hydraulic)	Nozzle Pressure: 0.2 – 2.5 bar Rope Speed: 150 – 500 m/min Loop Cycle Time: ≤ 50 sec	Max Temp: 135 - 140°C Heat Rate: 0.5 - 3.0°C/min Cool Rate: 1.5 - 2.0°C/min	Fabric creasing, rope marks, surface pilling, tangling
Package / Cone Dyeing	Spun Yarn & Filament Cones	1:5 – 1:8	Flow Rate: 35 – 50 L/kg/min Diff. Pressure (ΔP): 0.8 – 1.8 bar Density: 0.32 – 0.40 g/cm³	Max Temp: 135°C Heat Rate: 1.0 - 2.5°C/min Flow Reversal: 4 min In-Out / 2 min Out-In	Inner-to-outer cone shade variation, package channeling
Beam Dyeing	Open-width knits, tricot, lace	1:8 – 1:12	Max ΔP: 1.2 bar Uniform beam porosity across length	Max Temp: 130°C Heat Rate: 0.8 - 1.5°C/min	Moiré effect, edge-to-center shade variation (listing)
Continuous Pad-Steam / Thermosol	High-volume Woven Cotton & Blends	Pad-Liquor Trough (Continuous)	Wet Pick-up (WPU): 55 – 70% Mangle Pressure: 1.5 – 4.0 bar Steamer Dwell: 60 – 90 sec	Thermosol Temp: 200 – 215°C Dwell Time: 45 – 60 sec Tolerance: ±1.5°C width-wise	Intermediate migration, center-to-selvedge tailing

3. Core Mathematical Calculations for Wet Processing

- 1. Total Bath Volume (L): $V_{bath} = \text{Dry Substrate Mass (kg)} \times \text{MLR Ratio Factor}$
- 2. Dye Quantity Calculation (g): $\text{Dye Mass} = [\text{Shade \%} \times \text{Dry Fabric Mass (kg)} \times 1000] / 100$
- 3. Auxiliary Chemical Mass (g): $\text{Chemical Mass} = \text{Dosage (g/L)} \times V_{bath} \text{ (Liters)}$
- 4. Pump Turnover Rate Constraint: $\text{Turnover Time } T_{turn} = V_{bath} \text{ (Liters)} / Q_{pump} \text{ (L/min)}$ [Target: $T_{turn} \leq 1.5 \text{ min}$]
- 5. Rope Loop Cycle Time: $T_{rope} = [\text{Loop Length (m)} / \text{Rope Speed (m/min)}] \times 60$ [Target: $45 \leq T_{rope} \leq 60 \text{ seconds}$]

4. Standard Industrial Dyeing Recipes & Thermal Curves

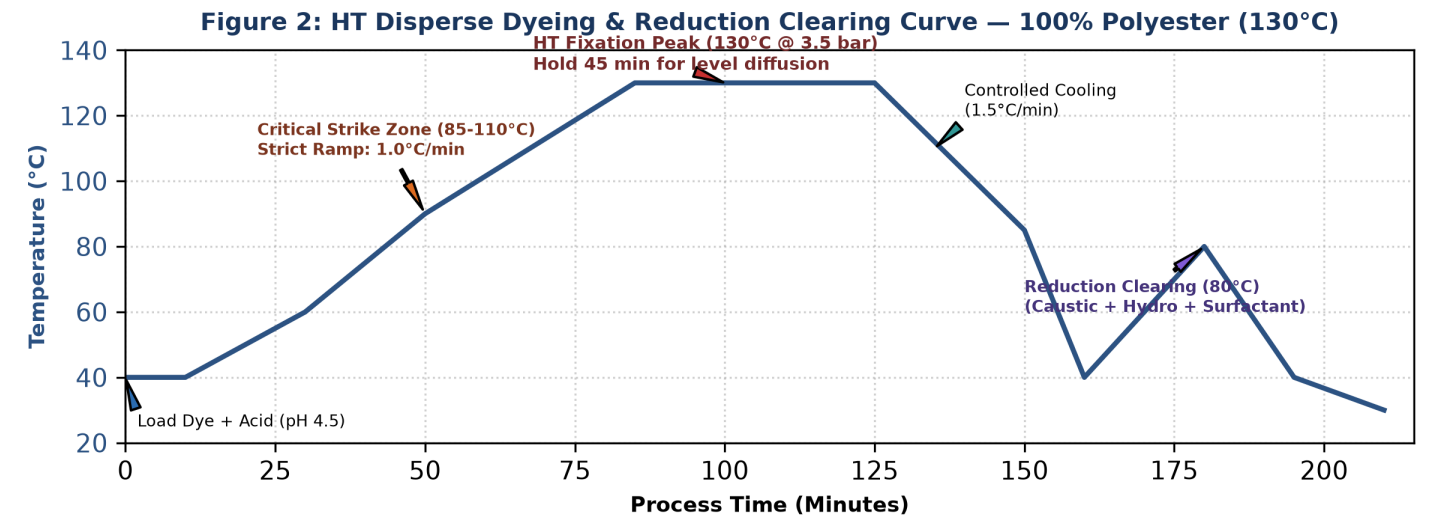
Figure 1: Standard Exhaust Dyeing Curve — 100% Cotton Reactive Dye (HE/Hot-Brand)



Recipe Formulation 1: 100% Combed Cotton Knit — Reactive Exhaust Dyeing (60°C)

Component / Auxiliary	Chemical Role / Function	Light Shade (<1.0%)	Medium Shade (1-2%)	Dark Shade (2-4%)	Black / Deep Dark (>4%)
Bifunctional Reactive Dye	Colorant (Reactive vinylsulfone/monochlorotriazine)	0.50 % owf	1.50 % owf	3.00 % owf	6.50 % owf
Sequestering Agent	Water softening / metal ion chelation (Fe, Ca, Mg)	1.0 g/L	1.0 g/L	1.2 g/L	1.5 g/L
Wetting / Leveling Agent	Surface tension reduction & uniform migration	1.0 g/L	1.0 g/L	1.0 g/L	1.0 g/L
Anti-Creasing Lubricant	Internal fabric rope lubrication	1.0 g/L	1.5 g/L	1.5 g/L	2.0 g/L
Glauber's Salt (Na ₂ SO ₄)	Electrolyte for dye exhaustion promotion	20 g/L	40 g/L	60 g/L	80 g/L
Soda Ash (Na ₂ CO ₃)	Alkali for fixation reaction (Target pH 10.8 - 11.2)	10 g/L	15 g/L	20 g/L	20 g/L
Caustic Soda (38°Bé)	Alkali booster (only for deep dark/black)	0.0 g/L	0.0 g/L	0.0 g/L	1.5 g/L

High-Temperature Disperse Dyeing & Reduction Clearing



Recipe Formulation 2: 100% Polyester (PES) — HT Exhaust Dyeing (130°C)

Process Stage	Chemical / Auxiliary	Industrial Dosage	Process Conditions & Critical Constraints
Main Dyeing Bath	HT Disperse Dye Acetic Acid (80%) Sodium Acetate Dispersing Agent Anti-Creasing Agent	0.1 – 4.0 % owf 1.2 g/L 0.8 g/L 1.5 g/L 1.0 g/L	pH target: 4.5 – 5.5 (Strict) Ramp rate between 85–110°C must be 1.0°C/min. Hold at 130°C & 3.5 bar for 45 minutes. Cool at 1.5°C/min to 85°C to avoid oligomers.
Reduction Clearing (RC) (For Medium/Dark Shades)	Caustic Soda (Flakes) Sodium Hydrosulfite (85%) Non-ionic Detergent	2.5 g/L 2.5 g/L 1.0 g/L	Run at 80°C for 20 minutes . Removes surface-adsorbed unfixed disperse dye. Essential for achieving Wash Fastness Grade 4-5 and Rubbing Fastness Grade 4.

5. Troubleshooting Matrix for Machine & Process Defects

Observed Defect	Root Cause Analysis (Machine & Process)	Corrective & Preventive Actions
Rope Marks / Running Creases	1. Nozzle pressure too low, causing fabric rope stall. 2. Rope speed < 150 m/min (loop time > 60 s). 3. Thermal shock from rapid cooling (> 2.5°C/min).	Increase nozzle pressure; synchronize reel speed; limit cooling rate to 1.5°C/min; add synthetic anti-crease lubricant.
Inner-to-Outer Cone Shade Variation	1. Yarn winding density too high (>0.40 g/cm³). 2. Pump flow rate < 35 L/kg/min. 3. Flow reversal cycle unbalanced (excessive In-to-Out).	Control soft winding density to 0.34 ± 0.02 g/cm³; set flow reversal to 4 min In-Out / 2 min Out-In; maintain ΔP at 1.2 bar.
Unlevel / Patchy Dye Strike	1. Alkali added rapidly in single dose (Reactive dye). 2. Temperature ramp too steep during exhaustion phase. 3. Bath turnover time > 1.5 min.	Implement progressive linear/exponential alkali dosing over 30 min; slow ramp to 1.0°C/min in exhaustion window; check main pump flow.
White Powder / Oligomer Specks	Cyclic oligomers migrating out of polyester fibers during HT dyeing at 130°C and crystallizing upon slow cooling.	Perform high-temperature drain at 85°C; perform reduction clearing with caustic soda and hydrosulfite at 80°C for 20 min.

6. Right-First-Time (RFT) Quality Control Protocol

- **Water Softening:** Maintain incoming process water hardness < 5 ppm CaCO₃ and Total Iron < 0.1 ppm.
- **Fabric Preparation:** Verify residual peroxide content < 10 ppm and core pH = 6.5 – 7.0 before reactive dye loading.
- **Instrument Calibration:** Calibrate PT100 temperature sensors (±0.5°C accuracy) and differential pressure transducers monthly.
- **Dosing Integrity:** Never add solid chemicals directly to the main vessel; pre-dissolve completely in auxiliary tank.